ROBERTO PRESTE

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Senior Full-Stack Engineer with 7+ years of experience developing production software end to end, from Python backends, REST APIs and SQL/PostgreSQL data layers to TypeScript web frontends. Builds and maintains traceable, well-documented applications for laboratory workflows in an ISO 13485-accredited, regulated life-sciences environment, where version control, logging, testing and quality documentation must stand up to audit. Strong in CI/CD, automated testing, AWS and Kubernetes, with experience in event-driven architecture, code review and peer mentoring, and a client-facing consulting background. Native Italian speaker, fluent in English.

PROFESSIONAL EXPERIENCE

Senior Software Engineer | Illumina

Oct 2021 - Oct 2026 | Cambridge, UK

- Develop end-to-end features for production full-stack applications, REST APIs and services supporting laboratory workflows in an ISO 13485-accredited, regulated environment, using Python, Flask, SQLAlchemy and SQL on the backend and TypeScript/Angular on the frontend.
- Designed and implemented an event-driven data integration architecture using AWS Lambda, Kafka, Snowflake and Terraform, replacing polling across three data sources and reducing data latency from 3–4 hours to near real time.
- Helped design and execute the migration of production workloads from Rancher Kubernetes to AWS EKS; accelerated the final migration during critical cluster instability, moving all remaining applications within one week with minimal downtime.
- Build and maintain CI/CD pipelines and cloud infrastructure using GitHub Actions, Docker, AWS, Kubernetes/EKS, Terraform and Helm.
- Adapt and extend applications for new laboratory workflows, translating user requirements into technical designs, prototypes and production-ready functionality.
- Write clean, tested code with automated testing (pytest) and contribute to software NCRs, CAPAs, root-cause investigations and technical documentation, keeping changes traceable under ISO 13485 quality processes; take part in code reviews and peer mentoring of engineers and interns, and support incident resolution, release planning and bug triage.

Scientific Software Developer | Enthought

Oct 2019 - Oct 2021 | Cambridge, UK

- Delivered client-facing consulting projects, designing and developing Python-based scientific desktop and web applications for data management, analysis and visualisation, using NumPy, Pandas, SciPy and Traits/TraitsUI across life-sciences and physics domains.
- Designed software for integrating and analysing data from heterogeneous sources, working directly with client researchers and data analysts to translate complex scientific workflows into maintainable technical solutions.
- Collaborated with engineers and project managers throughout architecture, implementation, testing and release, iteratively refining applications based on client feedback.

Bioinformatician | ELIXIR Italy

May 2019 - Jun 2019 | Bari, Italy

- Developed the “Implementation of HmtPhenome” ELIXIR project at the University of Bari.

Bioinformatician | ELIXIR Italy

May 2018 - Jun 2018 | Bari, Italy

- Developed the “Integration of the ELIXIR-IIB HmtDB resource into RD-Connect” ELIXIR project at the National Research Council.

Web Developer | ELIXIR Hub / EMBL-EBI

Mar 2016 - Sep 2016 | Cambridge, UK

- Implemented the Bioschemas ELIXIR website.

TECHNICAL SKILLS

Languages

Python · TypeScript/JavaScript · SQL · Shell · Go · R

Cloud & Infrastructure

CI/CD Pipelines · GitHub Actions · Docker · AWS · Kubernetes · Terraform · Helm · Ansible

Backend & Data

FastAPI · Flask · SQLAlchemy · REST APIs · PostgreSQL · MySQL · Kafka · Snowflake

Engineering

Full-stack Development · Automated Testing · Git & GitHub · Microservices · Event-driven Architecture · Infrastructure as Code · AI-assisted Development (Claude Code, Cursor)

Technical Leadership

Architecture & Design · Regulated Environments (ISO 13485) · Traceability & Quality Documentation · Production Operations · Incident Resolution · Peer Mentoring · Cross-team Engineering

EDUCATION

PhD - Functional and Applied Genomics and Proteomics

University of Bari “Aldo Moro” | 2016 - 2020

- Thesis: “Bioinformatics software and tools for Big Data in mitochondrial genomics”.
- Designed and developed bioinformatics software, databases and web services supporting analysis of mitochondrial genomes, variants, phenotypes and disease associations.

Master’s Degree - Cellular and Molecular Biology

University of Bari “Aldo Moro” | 2013 - 2015

Bachelor’s Degree - Medical and Pharmaceutical Biotechnologies

University of Bari “Aldo Moro” | 2009 - 2013

CERTIFICATIONS & TRAINING

AWS Certified Cloud Practitioner | Jul 2025
Security Engineering on AWS | Oct 2025
Running Containers on Amazon EKS | Mar 2025